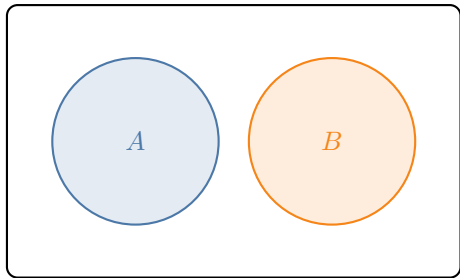


Mutually exclusive

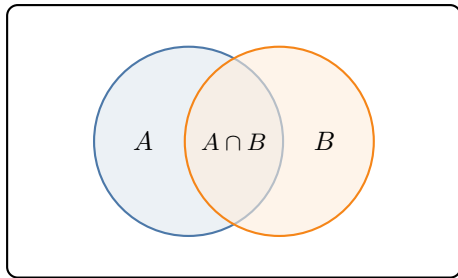


no overlap: $P(A \cap B) = 0$

cannot happen together

knowing B makes A impossible: $P(A \mid B) = 0$

Independent



overlap of just the right size: $P(A \cap B) = P(A)P(B)$

can happen together

knowing B changes nothing: $P(A \mid B) = P(A)$